

Factorial Experiments

1. A soft drink bottler is interested in obtaining more uniform fill height in the bottles produced by his manufacturing process the process engineer can control percentage carbonation, the operating pressure in the filler and bottles produced per minute or line speed during filling process. Two levels of carbonation, two levels of pressure and two levels of line speed are selected to study the effect of these factors in filling process. A factorial experiment two replicates is conducted, the order in which the 16 observations are taken is determined randomly. The coded data are given below. Analyze the data using Yates technique for computing contrast.

Percent Carbonation	Operating Pressure			
	Line Speed		Line Speed	
	200	250	200	250
10	-3	-1	-1	1
	-1	0	0	1
12	0	2	2	6
	1	1	3	5

2. Given the following ANOVA table:

S.O.V	d.f	Mean Square
A	2	5341.86
B	2	19558.36
AB	4	2403.44
Error	27	675.21
Total	35	

Assuming completely randomized model, interpret the effects of A, B and AB, when

- i) The levels of both factors are fixed
 - ii) The levels of both factors are random
 - iii) Levels of A are fixed but levels of B are random
3. The following data were obtained from a 2^3 factorial experiment; use yates method to analyze the data

Blocks	Treatment Combinations							
	(1)	a	b	c	ab	ac	bc	abc
I	12	14	15	14	14	10	11	11
II	21	19	16	17	18	16	18	15
III	10	12	10	14	11	11	10	12

4. An experiment was laid to see the effect of three fertilizers n, p and k at two levels each (i.e. applied and not applied). The eight combinations were randomized in each replication. Set up an analysis of variance to test the Main Effects and Interactions from the data given below:

Replicates	Treatments							
	(1)	n	p	np	k	nk	pk	npk
1	26	25	24	24	29	30	22	32
2	30	31	31	27	39	34	25	23
3	23	33	27	23	31	29	24	37
4	28	26	24	27	36	36	29	34

5. Given below are the treatment totals for different treatment combinations in RCB design with 3 replications

(1)	N	P	NP	K	NK	PK	NPK
100.7	117.2	103.8	144.3	100.9	106.2	98.5	142.8

The block totals are 303.1, 308.6 and 302.7. The total sums of squares are 1031.48. Complete ANOVA table and draw conclusions.

6. Eight treatments have been applied to a certain crop and their effects have been reflected in the following table

(1)	a	b	ab	c	ac	bc	abc
13	10	12	14	9	11	8	7
14	11	13	15	10	12	9	8

Give a method of analysis and interpret your results.

Split Plot Design

1. In a split plot design the whole plot treatment (A) was applied to 3 blocks and treatment (B) was applied to split plots. The total SS, Block SS, Whole Plot Error SS are 822.97, 77.55 and 36.28 respectively. The sums for 3 blocks are given in the following table:

	B ₁	B ₂	B ₃	B ₄
A ₁	89	104	118	117
A ₂	100	117	119	126
A ₃	92	92	104	121

Complete ANOVA table and draw conclusion. Also compute standard error for the

- (i) Difference between two 'A' treatment means
 - (ii) Difference between two 'B' treatment means
2. In an experiment on rice, there are 4 methods of irrigation to compared, on main plot, with 3 fertilizer mixtures on subplots. Two complete blocks were used. The Total SS, Block SS and Whole Plot Error are 3.0200, 0.0004 and 0.6509 respectively. The following table shows the sum of two blocks. Complete ANOVA table and draw conclusion.

	I ₁	I ₂	I ₃	I ₄
F ₁	4.68	4.34	3.78	4.67
F ₂	5.02	4.91	4.01	4.67
F ₃	6.00	5.16	4.10	5.45

- i) Find the standard error of difference between two Fertilizer means.
 - ii) Find the standard error of difference between two irrigation means.
3. Complete ANOVA table and interpret your results

S.O.V	d.f	SS	MS	F-ratio
Replicates	2	-	981.5	-
A	3	-	-	14
Error (i)	-	-	296	-
B	2	-	-	-
AB	-	2598	-	-
Error (ii)	-	-	621	-
Total	35	29271		

1. Complete ANOVA table and interpret your results.

S.O.V	d.f	SS	MS
Replicates	3	28.44	-
A	3	-	1.64
Linear	-	4.75	-
Non-linear	-	-	-
Error (i)	-	-	-
Subtotal	-	40.85	-
B	-	29.30	-
Linear	-	-	-
Non-linear	-	4.55	-
AB	9	-	-
Error (ii)	-	86.65	-
Total	-	206.69	-

Also compute standard error for the

- (iii) Difference between two 'A' treatment means
- (iv) Difference between two 'B' treatment means